

Linear algebra: Exercise Chapter 7

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1. Compute eigenvalues and eigenvectors of the following matrices.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} -3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -3 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 3 & 4 \\ 3 & 3 & 5 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 3 & -19 & 11 \\ 0 & 1 & 7 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Answer.

Matrix A:

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 1 - \lambda & 2 \\ 2 & 1 - \lambda \end{vmatrix} = 0 \\ \implies (1 - \lambda)(1 - \lambda) - 4 &= 0 \\ \implies \lambda_1 = -1, \lambda_2 = 3. \end{aligned}$$

Eigenvectors for $\lambda_1 = -1$:

$$(A - (-1)I)x = 0 \implies \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies \begin{cases} 2x_1 + 2x_2 = 0, \\ 2x_1 + 2x_2 = 0 \end{cases} \implies x_2 = -x_1.$$

Thus,

$$\begin{bmatrix} x_1 \\ -x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \implies v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Eigenvectors for $\lambda_2 = 3$:

$$(A - I)x = 0 \implies \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies \begin{cases} -2x_1 + 2x_2 = 0, \\ -2x_1 + 2x_2 = 0 \end{cases} \implies x_2 = x_1.$$

Thus,

$$\begin{bmatrix} x_1 \\ x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \implies v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Matrix B: This is a diagonal matrix, so the eigenvalues are the diagonal elements:

$$\lambda_1 = -3, \quad \lambda_2 = 5, \quad \lambda_3 = -3.$$

Eigenvectors for $\lambda_1 = -3$:

$$(B - (-3)I)x = 0 \implies \begin{bmatrix} 0 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies x_2 = 0.$$

Thus,

$$\begin{bmatrix} x_1 \\ 0 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \implies v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Eigenvectors for $\lambda_2 = 5$:

$$(A - 5I)x = 0 \implies \begin{bmatrix} -8 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies \begin{cases} x_1 = 0, \\ x_3 = 0. \end{cases}$$

Thus,

$$\begin{bmatrix} 0 \\ x_2 \\ 0 \end{bmatrix} = x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \implies v_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

Matrix C:

$$\begin{aligned} \det(C - \lambda I) &= \begin{vmatrix} 1 - \lambda & 0 & 1 \\ 2 & 3 - \lambda & 4 \\ 3 & 3 & 5 - \lambda \end{vmatrix} = (1 - \lambda) \begin{vmatrix} 3 - \lambda & 4 \\ 3 & 5 - \lambda \end{vmatrix} + 1 \begin{vmatrix} 2 & 3 - \lambda \\ 3 & 3 \end{vmatrix} \\ &= (1 - \lambda)(3 - \lambda)(5 - \lambda) - 12(1 - \lambda) + 6 - 3(3 - \lambda) \\ &= (1 - \lambda)(3 - \lambda)(5 - \lambda) - 15(1 - \lambda) \\ &= (1 - \lambda)\lambda(\lambda - 8). \end{aligned}$$

Thus,

$$\det(C - \lambda I) = 0 \implies \lambda(1 - \lambda)(\lambda - 8) = 0 \implies \lambda_1 = 0, \lambda_2 = 1, \lambda_3 = 8.$$

Eigenvectors for $\lambda_1 = 0$:

$$(C - 0I)x = 0 \implies Cx = 0 \implies \begin{bmatrix} 1 & 0 & 1 \\ 2 & 3 & 4 \\ 3 & 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies \begin{cases} x_1 + x_3 = 0, \\ 2x_1 + 3x_2 + 4x_3 = 0, \\ 3x_1 + 3x_2 + 5x_3 = 0 \end{cases} \implies x_3 = -x_1, x_2 = \frac{2}{3}x_1.$$

Thus,

$$\begin{bmatrix} x_1 \\ \frac{2}{3}x_1 \\ -x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ \frac{2}{3} \\ -1 \end{bmatrix} \implies v_1 = \begin{bmatrix} 1 \\ \frac{2}{3} \\ -1 \end{bmatrix}.$$

Eigenvectors for $\lambda_2 = 1$:

$$(C - I)x = 0 \implies \begin{bmatrix} 0 & 0 & 1 \\ 2 & 2 & 4 \\ 3 & 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies \begin{cases} x_3 = 0, \\ 2x_1 + 2x_2 + 4x_3 = 0, \\ 3x_1 + 3x_2 + 4x_3 = 0 \end{cases} \implies x_3 = 0, x_2 = -x_1.$$

Thus,

$$\begin{bmatrix} x_1 \\ -x_1 \\ 0 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \implies v_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}.$$

Eigenvectors for $\lambda_3 = 8$:

$$(C - 8I)x = 0 \implies \begin{bmatrix} -7 & 0 & 1 \\ 2 & -5 & 4 \\ 3 & 3 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies \begin{cases} -7x_1 + x_3 = 0, \\ 2x_1 - 5x_2 + 4x_3 = 0, \\ 3x_1 + 3x_2 - 3x_3 = 0 \end{cases} \implies x_3 = 7x_1, x_2 = 6x_1.$$

Thus,

$$\begin{bmatrix} x_1 \\ 6x_1 \\ 7x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 6 \\ 7 \end{bmatrix} \implies v_3 = \begin{bmatrix} 1 \\ 6 \\ 7 \end{bmatrix}.$$

Matrix D: Since this is an upper triangular matrix, the eigenvalues are the diagonal elements:

$$\lambda_1 = 2, \quad \lambda_2 = 1.$$

Eigenvectors for $\lambda_1 = 2$:

$$(D - 2I)x = 0 \implies \begin{bmatrix} 0 & 3 & -19 & 11 \\ 0 & -1 & 7 & 0 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \implies \begin{cases} 3x_2 - 19x_3 + 11x_4 = 0, \\ -x_2 + 7x_3 = 0, \\ -x_3 - x_4 = 0, \\ -x_4 = 0 \end{cases} \implies x_2 = x_3 = x_4 = 0.$$

Thus,

$$\begin{bmatrix} x_1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \implies v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Eigenvectors for $\lambda_1 = 1$:

$$(D - I)x = 0 \implies \begin{bmatrix} 1 & 3 & -19 & 11 \\ 0 & 0 & 7 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \implies \begin{cases} x_1 + 3x_2 - 19x_3 + 11x_4 = 0, \\ 7x_3 = 0, \\ -x_4 = 0 \end{cases} \implies x_1 = -3x_2, x_3 = x_4 = 0.$$

Thus,

$$\begin{bmatrix} x_2 \\ -3x_2 \\ 0 \\ 0 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ -3 \\ 0 \\ 0 \end{bmatrix} \implies v_2 = \begin{bmatrix} 1 \\ -3 \\ 0 \\ 0 \end{bmatrix}.$$

2. Let $2, -3, 2, 5$ are the eigenvalues of 4×4 matrix A . Find the determinant and trace of A .

Answer.

$$|A| = 2 \times (-3) \times 2 \times 5 = -60, \quad \text{tr}(A) = 2 - 3 + 2 + 5 = 6.$$

3. Let λ is the eigenvalue of $n \times n$ matrix A corresponding the eigenvector x . Prove the following statements:

- (a) λ^2 is an eigenvalue of A^2 corresponding the eigenvector x .
- (b) If A is invertible, then λ^{-1} is an eigenvalue of A^{-1} corresponding the eigenvector x .
- (c) For each $\mu \in \mathbb{R}$, $\lambda + \mu$ is an eigenvalue of $A + \mu I$ corresponding the eigenvector x .
- (d) For each $\mu \in \mathbb{R}$, $\mu\lambda$ is an eigenvalue of μA corresponding the eigenvector x .

Answer. Since λ is the eigenvalue of A corresponding the eigenvector x , $Ax = \lambda x$.

Part (a):

$$A^2x = AAx = A(\lambda x) = \lambda Ax = \lambda^2 x.$$

Part (b):

$$Ax = \lambda x \implies A^{-1}Ax = A^{-1}\lambda x \implies \frac{1}{\lambda}x = A^{-1}x.$$

Part (c):

$$(A + \mu I)x = Ax + \mu Ix = \lambda x + \mu x = (\lambda + \mu)x.$$

Part (d):

$$(\mu A)x = \mu Ax = \mu \lambda x = (\mu \lambda)x.$$

4. Consider the matrix A in question 1. Compute the eigenvalues and eigenvectors of matrices $A - 3I$, $-2A$, A^{-1} , A^2 .

Answer.

- (a) $\lambda_1 = -11$ and $\lambda_2 = -2$ are the eigenvalues of $A - 3I$.
- (b) $\lambda_1 = 16$ and $\lambda_2 = -2$ are the eigenvalues of $-2A$.
- (c) $\lambda_1 = -\frac{1}{8}$ and $\lambda_2 = 1$ are the eigenvalues of A^{-1} .
- (d) $\lambda_1 = 64$ and $\lambda_2 = 1$ are the eigenvalues of A^2 .

The eigenvector corresponding each eigenvalue does not change.

5. Compute A^5x for the following matrix A and vector x :

$$A = \begin{bmatrix} 2 & 4 & -2 \\ 0 & 10 & -4 \\ 0.5 & 12 & -4 \end{bmatrix}, \quad x = \begin{bmatrix} 2 \\ 4 \\ 7 \end{bmatrix}.$$

Answer.

$$Ax = \begin{bmatrix} 2 & 4 & -2 \\ 0 & 10 & -4 \\ 0.5 & 12 & -4 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 7 \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \\ 21 \end{bmatrix} = 3x \implies A^5x = 3^5x.$$